Credit Card Fraud Detection & Anomaly Audit Report

Production Intelligence Report | Generated: October 01, 2026 - 06:40:10 UTC | Strict Compliance

1. Executive Transaction Summary

Total Scored	**568,654**	**Flagged / Anomalous**	**12,768**
Anomaly Rate	**0.29%**	**Avg Transaction**	**$88.51**
Active Model	**Isolation Forest**	**Model Version**	**v1.0.0**

2. Active Model Performance & Benchmark Metrics

Metric	Score	Benchmark Guideline	Operational Status
Precision	1.0000	Minimizes false customer card declines	**Optimal**
Recall (Sensitivity)	0.8833	Maximizes fraud dollar capture	**Optimal**
F1-Score	0.9381	Harmonic balance of Precision & Recall	**Robust**
PR-AUC (Avg Precision)	0.9905	Primary metric for extreme imbalance	**High**
ROC-AUC	0.9996	Overall ranking capability across thresholds	**Superior**
False Positive Rate (FPR)	0.0000	Under 2% friction on genuine cardholders	**Within SLA**

3. Continuous Learning & Data Drift Diagnostics

Diagnostic Check	Observed Value	Threshold SLA	Automated Action
Population Stability (PSI)	0.0520	< 0.25 (Standard Threshold)	**STABLE**
Retraining Mode	Batch & Incremental Trigger	Continuous Drift Sentinel	**Enabled**
Last Retraining Event	Baseline Deployment	Scheduled / Drift Triggered	**SUCCESS**

4. Sample Flagged Transactions & Root-Cause Explainability

TX ID	Amount	Merchant / Channel	Score	Classification	Contributing Risk Factors
TX-383721	$0.77	Unknown (Card)	**0.500**	**Suspicious**	["Unusual transaction time (23:00 hours)", "V
TX-383714	$9.87	Unknown (Card)	**0.471**	**Suspicious**	["Unusual transaction time (23:00 hours)", "V
TX-383654	$381.05	Unknown (Card)	**0.483**	**Suspicious**	["Unusual transaction time (23:00 hours)", "A
TX-383652	$405.09	Unknown (Card)	**0.474**	**Suspicious**	["Unusual transaction time (23:00 hours)", "A
TX-383651	$664.60	Unknown (Card)	**0.572**	**Suspicious**	["Unusual transaction time (23:00 hours)", "A

TX-383631	$787.95	Unknown (Card)	**0.589**	**Suspicious**	["Unusual transaction time (23:00 hours)", "A